Graph Neural Networks and Graph Foundation Models for Scientific Discovery: Architectures, Paradigms, and Frontiers

1 Introduction

The integration of graph-based deep learning into scientific inquiry marks a paradigmatic shift from data-driven pattern recognition to physics-informed discovery, fundamentally accelerating the design-make-test-analyze cycle in chemistry, materials science, and biology. Unlike social or information networks where relationships are often abstract, scientific graphs encode rigid geometric constraints, conservation laws, and multi-scale hierarchies that govern physical reality. Early approaches, such as Gated Graph Sequence Neural Networks [1] and standard Message Passing Neural Networks, established the utility of learning from structured data but often lacked the inductive biases necessary to respect the symmetries of three-dimensional space. Consequently, the field is rapidly transitioning from shallow, task-specific architectures toward generalizable Graph Foundation Models capable of zero-shot transfer across disparate scientific domains [2].

The distinctive nature of scientific data necessitates architectural innovations that transcend standard benchmark evaluations. Scientific graphs are not merely topological; they are geometric embeddings where node positions and edge attributes must adhere to rotational, translational, and permutational equivariance. Traditional models often fail to capture these properties, leading to poor generalization in out-of-distribution regimes common in experimental science. Recent advances in geometric deep learning have addressed this by incorporating steerable features and equivariant message passing, ensuring that predictions remain consistent under physical transformations [3; 4]. Furthermore, the expressive power of these models is being pushed beyond the limits of the Weisfeiler-Lehman test through subgraph counting and higher-order tensor operations, enabling the discrimination of complex molecular topologies and crystal lattices that were previously indistinguishable [5; 6].

This survey delineates the evolution from specialized models to foundation models, highlighting how self-supervised pre-training on massive corpora enables the capture of universal chemical and physical principles. While early methods like Graph Convolutional Policy Networks focused on goal-directed generation within narrow constraints [7], modern foundation models leverage unified tokenization and scalable transformers to process heterogeneous graphs ranging from small molecules to extended solid-state materials [8; 9]. We critically analyze the trade-offs between strict equivariance and computational efficiency, as well as the challenges of integrating multimodal data, including textual scientific literature and spectral observations, into cohesive reasoning frameworks [10].

The scope of this review encompasses theoretical expressivity, architectural paradigms for equivariance, self-supervised pre-training strategies, generative models for autonomous discovery, and mechanisms for trustworthiness and causal reasoning. We synthesize insights from over a decade of research, contrasting the limitations of local message passing with the global context capture of graph transformers [11; 12]. By establishing a rigorous taxonomy of methods and evaluating their impact on high-stakes applications such as drug discovery and catalyst design [13; 14], this work provides a roadmap for the next generation of AI-driven scientific instruments. Ultimately, we argue that the convergence of neuro-symbolic reasoning, physics-informed constraints, and large-scale pre-training will define the future of automated scientific discovery, bridging the gap between associative learning and causal understanding.

2 Theoretical Foundations and Expressive Architectures for Scientific Data

2.1 Theoretical Expressivity and Structural Discrimination Limits

The representational capacity of Graph Neural Networks (GNNs) in scientific discovery is fundamentally constrained by their inability to distinguish non-isomorphic graph structures that share identical local neighborhoods. Standard Message Passing Neural Networks (MPNNs) operate within the expressive bounds of the 1-dimensional Weisfeiler-Lehman (1-WL) test, rendering them incapable of differentiating certain complex molecular topologies or crystal lattices that are critical for property prediction [15; 16]. This theoretical limitation is particularly acute in scientific domains where subtle structural variations, such as the distinction between specific ring systems or stereoisomers, dictate physicochemical behavior. Recent theoretical analyses have further exposed that standard MPNNs suffer from exponential information loss as depth increases, often converging to representations that capture only connected components and node degrees, thereby failing to propagate long-range dependencies essential for modeling extended materials or large biomolecules [17].

To transcend the 1-WL horizon, architectural innovations have focused on enhancing substructure awareness and higher-order interactions. A prominent direction involves explicitly encoding subgraph counts or motifs, which are invariant features invisible to standard aggregation schemes. Models such as Graph Substructure Networks (GSN) augment message passing with handcrafted substructure identifiers, achieving expressivity strictly superior to 1-WL while maintaining linear complexity [18]. However, the counting power of these models remains bounded; for instance, standard Subgraph MPNNs fail to count cycles larger than four nodes, a significant deficit for identifying benzene rings or larger macrocycles in organic chemistry [6]. Advanced variants like I^2-GNNs address this by assigning unique identifiers to root nodes and their neighbors within subgraphs, theoretically enabling the counting of up to 6-cycles and offering a practical balance between discriminative power and computational efficiency. Alternatively, Relational Pooling (RP) frameworks leverage finite partial exchangeability to construct maximal representations, effectively surpassing the WL hierarchy by incorporating global permutation-invariant operations [19].

Beyond subgraph enumeration, the integration of higher-order tensor operations offers a path toward universality equivalent to the k-WL test. While k-order invariant networks can distinguish graphs indistinguishable by lower-order tests, they incur prohibitive computational costs scaling exponentially with k [20]. To mitigate this, sparse and localized higher-order architectures have been proposed, which consider subsets of vertex tuples to achieve expressivity beyond 1-WL without the full overhead of global tensor operations [21]. Furthermore, the injection of structural noise or unique node identifiers has been shown to break symmetries that trap standard MPNNs, allowing them to approximate the expressive power of more complex architectures through simplified mechanisms [22].

Despite these advances, challenges persist in generalizing learned representations across varying graph sizes and distributions, a common requirement in scientific extrapolation. Theoretical work indicates that when local structures depend on global graph size, standard GNNs may converge to non-generalizing solutions, necessitating self-supervised objectives that explicitly enforce size-invariant structural learning [23]. Future research must therefore focus on synthesizing these higher-order inductive biases with scalable attention mechanisms and geometric constraints to develop architectures that are not only theoretically universal but also robust to the distributional shifts inherent in experimental scientific data.

2.2 Geometric Equivariance and Symmetry-Driven Inductive Biases

Addressing the gap identified in the previous discussion on topological expressivity, the modeling of scientific systems fundamentally requires adherence to the physical symmetries of Euclidean space, specifically invariance to translations and rotations, and equivariance to particle permutations. While earlier architectural innovations enhanced the ability to distinguish complex graph structures, they often treated nodes as abstract entities, neglecting the geometric coordinates and physical symmetries inherent in scientific systems. Standard message-passing neural networks frequently fail to strictly enforce these constraints, leading to data inefficiency and poor generalization in out-of-distribution regimes. To rectify this, Geometric Graph Neural Networks (GGNNs) have emerged as a critical architectural paradigm, explicitly encoding these symmetries as inductive biases. A primary dichotomy exists between invariant architectures, which map geometric inputs to scalar outputs, and equivariant architectures, which preserve the transformation properties of vector and tensor features throughout the network depth.

Strictly equivariant models operating in Cartesian bases, such as E(n)-Equivariant Graph Neural Networks (EGNNs), achieve rotation and translation equivariance by updating node coordinates directly alongside features, avoiding the computational burden of higher-order tensor representations [24]. These models have demonstrated remarkable efficacy in dynamical systems modeling and molecular property prediction. However, for tasks requiring the resolution of complex angular dependencies or higher-order geometric motifs, models utilizing spherical harmonics and irreducible representations offer superior expressivity. Steerable E(3) Equivariant Graph Neural Networks (SEGNNs) generalize this approach by allowing node attributes to be covariant vectors or tensors, thereby incorporating geometric and physical quantities like force and spin directly into the message-passing mechanism [3]. Theoretical analyses using the Geometric Weisfeiler-Lehman (GWL) test confirm that while invariant layers are limited in distinguishing one-hop identical geometric graphs, equivariant layers significantly expand the class of discriminable structures by propagating geometric information beyond local neighborhoods [5].

Despite the theoretical guarantees of strict equivariance, recent innovations suggest that relaxing these constraints can yield practical benefits in specific scientific contexts, bridging the gap toward the many-body interactions discussed subsequently. For instance, in systems where symmetries are partially broken by external fields such as gravity or electric potentials, enforcing full E(3) equivariance may be overly restrictive. Subequivariant Graph Neural Networks address this by relaxing continuous equivariance to subequivariance, allowing the model to learn dynamics in bounded or anisotropic environments while maintaining universal approximation capabilities [25]. Similarly, Discrete Equivariant Graph Neural Networks (DEGNNs) guarantee equivariance only to specific discrete point groups, enabling the use of richer geometric feature combinations that approximate unobserved interaction functions more effectively than their continuous counterparts [26]. Furthermore, approaches like ForceNet challenge the necessity of hard-coded symmetry constraints, demonstrating that physical constraints can be implicitly learned through physics-based data augmentation, achieving competitive accuracy with reduced computational overhead [27].

The choice between strict and relaxed symmetry enforcement involves a trade-off between data efficiency and model flexibility, setting the stage for more complex interaction modeling. While strict equivariance ensures adherence to conservation laws and rapid convergence in low-data regimes, relaxed or learned symmetry approaches offer greater adaptability to complex, real-world boundary conditions and broken symmetries. Emergent trends indicate a shift toward hybrid architectures, such as those employing virtual nodes to model hidden geometric entities like binding pockets, which enhance performance in tasks like protein-ligand identification without compromising the underlying equivariant framework [28]. Future research must further elucidate the conditions under which symmetry breaking improves generalization and develop unified frameworks that dynamically adjust equivariance constraints based on the local physical context. Building upon these geometric foundations, the subsequent section addresses a second critical limitation: the reliance of even geometric architectures on dyadic interactions, which proves insufficient for capturing the many-body potentials and long-range forces that dictate system behavior in quantum chemistry and condensed matter physics.

2.3 Capturing Higher-Order Interactions and Non-Local Dependencies

Standard message-passing neural networks (MPNNs) operate primarily on dyadic interactions, a limitation that proves critical in scientific domains where many-body potentials and long-range forces dictate system behavior. In quantum chemistry and condensed matter physics, properties often emerge from three-body angles, four-body torsions, and non-local electrostatic interactions that pairwise graphs fail to capture explicitly. To address this expressivity gap, recent architectural innovations have extended beyond simple edge-based aggregation to model higher-order topological structures and global dependencies.

Hypergraph formulations provide a natural framework for encoding polyadic relations, where a single hyperedge can connect multiple nodes to represent complex interaction motifs such as functional groups or crystal coordination polyhedra. While traditional hypergraph neural networks offer theoretical flexibility, they often struggle with scalability and heterogeneous hyperedge sizes. [29] addresses these challenges by introducing a self-attention mechanism capable of handling variable-sized hyperedges in both homogeneous and heterogeneous settings, significantly improving performance in genomics and network analysis. Furthermore, [30] demonstrates that converting hypergraphs into weighted clique expansions can allow standard GCNs to approximate higher-order learning efficiently, though this may incur memory overhead. Theoretical analyses suggest that while many HyperGNNs can be approximated by GNNs on expanded graphs [31], explicit higher-order modeling remains essential for tasks requiring precise substructure counting, which standard MPNNs and even 2-WL tests cannot perform for induced subgraphs of size three or greater [32].

To overcome the locality constraints of message passing, hierarchical pooling mechanisms construct multi-scale representations that capture collective modes and macroscopic phases. By adaptively coarsening graphs into super-nodes, models like [33] and [34] enlarge the receptive field, enabling the capture of global topological signatures without quadratic computational complexity. This hierarchical approach is particularly vital for modeling systems with distinct spatial scales, such as protein-ligand complexes or porous materials. Complementing this, [35] introduce efficient memory layers that jointly learn node representations and graph coarsening, achieving state-of-the-art results in graph classification by preserving long-range structural information.

Alternatively, transformer-inspired architectures leverage global self-attention to facilitate direct information flow between distant nodes, effectively bypassing the over-squashing bottleneck inherent in deep MPNNs. [36] proposes edge-augmented graph transformers that replace localized convolution with global attention, setting new benchmarks in quantum-chemical regression by capturing long-range dependencies critical for accurate property prediction. Similarly, [37] generalize message passing to shortest-path neighborhoods, allowing non-adjacent nodes to communicate directly and mitigating information loss during propagation. These global strategies are further enhanced by spectral methods; [38] introduces adaptive polynomial bases to handle heterophily and over-squashing, providing a robust mechanism for learning invariant representations across varying graph structures.

Despite these advances, trade-offs between expressive power, computational efficiency, and physical consistency persist. While higher-order tensor operations and global attention improve expressivity, they often scale poorly with system size, necessitating sparse approximations or localized variants [21]. Moreover, ensuring that these complex architectures respect physical symmetries and conservation laws remains an open challenge. Future directions likely lie in hybrid models that combine the locality of equivariant message passing with the global context of attention mechanisms, alongside neuro-symbolic approaches that explicitly encode physical laws into higher-order interaction terms to ensure robust generalization in out-of-distribution scientific regimes.

2.4 Continuous Dynamics, Neural Operators, and Differential Equations

Bridging the gap between discrete higher-order architectures and the continuous nature of physical laws, the integration of Graph Neural Networks (GNNs) with continuous mathematical frameworks represents a paradigm shift from fixed message-passing steps to the modeling of continuous-time dynamics and infinite-dimensional function spaces. This transition directly addresses the limitations of discrete solvers discussed previously, offering a principled approach for scientific domains where systems evolve continuously under differential equations. By formulating node representation updates as derivatives within Ordinary Differential Equations (ODEs), Graph Neural ODEs (GNN-ODEs) enable the simulation of complex particle dynamics with adaptive temporal resolution, effectively handling irregularly sampled data and mitigating the error accumulation inherent in discrete solvers [39]. Such continuous-depth models approximate the vector field of the system state rather than a fixed-step transition map, providing a natural extension to the dynamic formulations required for high-fidelity scientific modeling.

Recent advancements have extended this continuous formalism to rigorously incorporate the higher-order physical laws and symmetry constraints essential for physical consistency. For instance, the Second-order Equivariant Graph Neural Ordinary Differential Equation (SEGNO) explicitly embeds second-order motion laws and continuity into the architecture, ensuring that the learned trajectories between states are unique and physically plausible while maintaining equivariance [40]. Similarly, Lagrangian Graph Neural Networks (LGnn) provide a strong inductive bias by learning the system's Lagrangian directly from trajectories, simplifying the modeling of constrained systems and drag forces while achieving superior data efficiency compared to feed-forward alternatives [41]. These approaches demonstrate that embedding physical priors—such as the decoupling of kinetic and potential energies or the enforcement of conservation laws—significantly enhances rollout accuracy and generalizability to unseen system sizes [39], thereby resolving the trade-offs between expressivity and physical fidelity highlighted in prior sections.

Beyond temporal evolution, the concept of neural operators has emerged as a powerful tool for solving Partial Differential Equations (PDEs) on irregular domains, effectively scaling the continuous perspective to infinite-dimensional function spaces. Unlike traditional solvers that discretize space and time, graph-based neural operators learn mappings that serve as resolution-independent surrogates for computationally expensive simulations. The MeshGraphNets framework exemplifies this capability by adapting mesh discretization during forward simulation to accurately predict dynamics in aerodynamics and structural mechanics, achieving speedups of one to two orders of magnitude over conventional solvers [42]. Furthermore, hierarchical adaptive graph networks, such as GNN-Surrogate, facilitate efficient parameter space exploration for unstructured-mesh ocean simulations by predicting output fields across varying resolutions [43].

The synthesis of physics-informed constraints with these continuous graph learning frameworks further strengthens model robustness in data-scarce regimes. Physics-Informed Graph Networks (PIGNs) utilize graph exterior calculus to construct differential operators, enabling the solution of forward and inverse problems by integrating multimodal data with mathematical laws [44]. In spatial-temporal forecasting, customized graph convolution layers that honor physics-based constraints have demonstrated improved interpretability and robustness over purely data-driven baselines [45]. Looking forward, the convergence of continuous dynamics, neural operators, and symbolic regression offers a promising avenue for discovering interpretable physical laws from data, bridging the gap between black-box approximation and explicit scientific theory [46; 47]. Future research must address the scalability of these continuous formulations to trillion-parameter regimes and enhance their ability to generalize across disparate physical domains without retraining, laying the groundwork for the next generation of graph foundation models.

3 Graph Foundation Models and Self-Supervised Pre-training Strategies

3.1 Self-Supervised Pre-training Objectives and Geometric Consistency

The transition toward graph foundation models in scientific discovery necessitates pretext tasks that transcend generic topological augmentation, instead embedding the fundamental physical and chemical priors inherent to atomic systems. Traditional self-supervised approaches, such as node masking or edge prediction, often fail to capture the causal mechanisms governing molecular properties because they treat graphs as abstract combinatorial structures rather than geometric entities bound by physical laws. To address this, recent advancements have pivoted toward multi-view geometric contrastive learning, which explicitly leverages the consistency between 2D topological graphs and their 3D geometric conformers. Frameworks like GeomGCL [48] utilize dual-view message passing to align representations across dimensionalities, ensuring that learned embeddings remain robust even when 3D coordinates are unavailable during downstream deployment. This geometric consistency is critical, as the spatial arrangement of atoms dictates intermolecular forces and reactivity, aspects often lost in purely topological pre-training [24].

Beyond geometric alignment, there is a growing consensus that pretext tasks must operate at the scale of chemical motifs rather than individual atoms to capture higher-order semantic structures. Standard masked graph modeling, while effective in natural language processing, can violate chemical valency rules when applied naively to atomic graphs. Consequently, motif-centric reconstruction strategies have emerged, where models are tasked with predicting functional groups or BRICS fragments instead of isolated nodes. The Motif-based Graph Self-supervised Learning (MGSSL) framework [49] exemplifies this shift, employing a generative objective that reconstructs chemically valid subgraphs, thereby forcing the encoder to internalize the rules of chemical synthesis and stability. Similarly, recent analyses of masked graph modeling suggest that the choice of tokenizer—specifically moving from atom-level to subgraph-level tokens—combined with an expressive decoder, significantly enhances the quality of learned representations [50]. These approaches ensure that the latent space respects the hierarchical nature of chemistry, where local functional groups determine global physicochemical behavior.

Furthermore, the efficacy of these objectives is heavily dependent on domain-knowledge-guided augmentation strategies that prevent the learning of spurious correlations. Generic perturbations, such as random edge dropping, can alter the fundamental electronic properties of a molecule, leading to negative transfer. In contrast, frameworks like MoCL [51] incorporate chemical semantics into the augmentation process, ensuring that variations preserve graph semantics while introducing sufficient diversity for robust contrastive learning. This is complemented by global-semantic learning mechanisms, such as GraphLoG [52], which utilize hierarchical prototypes to capture cluster-level structures across the entire dataset, addressing the limitation of methods that focus solely on local instance discrimination. Despite these advances, challenges remain in scaling these geometrically consistent objectives to billion-parameter regimes without incurring prohibitive computational costs associated with 3D equivariant operations. Future directions must therefore explore efficient approximations of geometric consistency and the integration of neuro-symbolic constraints to further align pre-training objectives with the rigorous demands of scientific simulation and discovery.

3.2 Architectural Paradigms for Scalable Graph Foundation Models

While architectural innovations provide the structural backbone for Graph Foundation Models (GFMs), their efficacy in scientific discovery is ultimately governed by pretext tasks that transcend generic topological augmentation to embed fundamental physical and chemical priors. Traditional self-supervised approaches, such as node masking or edge prediction, often fail to capture the causal mechanisms governing molecular properties because they treat graphs as abstract combinatorial structures rather than geometric entities bound by physical laws. To address this limitation, recent advancements have pivoted toward multi-view geometric contrastive learning, which explicitly leverages the consistency between 2D topological graphs and their 3D geometric conformers. Frameworks like GeomGCL [48] utilize dual-view message passing to align representations across dimensionalities, ensuring that learned embeddings remain robust even when 3D coordinates are unavailable during downstream deployment. This geometric consistency is critical, as the spatial arrangement of atoms dictates intermolecular forces and reactivity, aspects often lost in purely topological pre-training [24].

Beyond geometric alignment, there is a growing consensus that pretext tasks must operate at the scale of chemical motifs rather than individual atoms to capture higher-order semantic structures compatible with the unified tokenization strategies discussed previously. Standard masked graph modeling, while effective in natural language processing, can violate chemical valency rules when applied naively to atomic graphs. Consequently, motif-centric reconstruction strategies have emerged, where models are tasked with predicting functional groups or BRICS fragments instead of isolated nodes. The Motif-based Graph Self-supervised Learning (MGSSL) framework [49] exemplifies this shift, employing a generative objective that reconstructs chemically valid subgraphs, thereby forcing the encoder to internalize the rules of chemical synthesis and stability. Similarly, recent analyses of masked graph modeling suggest that the choice of tokenizer—specifically moving from atom-level to subgraph-level tokens—combined with an expressive decoder, significantly enhances the quality of learned representations [50]. These approaches ensure that the latent space respects the hierarchical nature of chemistry, where local functional groups determine global physicochemical behavior, thus facilitating the transferability required for foundation models.

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instance discrimination. Despite these advances, challenges remain in scaling these geometrically consistent objectives to billion-parameter regimes without incurring prohibitive computational costs associated with 3D equivariant operations—a bottleneck that directly motivates the architectural shifts toward efficient attention and hybrid designs explored in the subsequent section. Future directions must therefore explore efficient approximations of geometric consistency and the integration of neuro-symbolic constraints to further align pre-training objectives with the rigorous demands of scientific simulation and discovery.

The transition from task-specific message-passing neural networks (MPNNs) to scalable Graph Foundation Models (GFMs) necessitates a fundamental reimagining of architectural inductive biases to accommodate the heterogeneity and scale of scientific data, directly addressing the computational bottlenecks identified in prior pretext design. Traditional MPNNs, while effective for localized property prediction, often struggle with over-smoothing and limited receptive fields when scaled to the billion-parameter regimes required for foundation modeling [9]. Consequently, emerging paradigms prioritize unified tokenization strategies and global attention mechanisms to capture long-range dependencies inherent in complex biomolecules and extended crystal lattices. A critical innovation in this domain is the construction of a standardized "graph vocabulary," which discretizes continuous scientific structures into transferable tokens, such as recurring molecular motifs or lattice patterns [53]. This approach, exemplified by unified graph tokenizers, enables models to generalize across unseen graph topologies and domains by adapting to diverse structural properties without retraining [8], thereby providing the structural substrate necessary for the sophisticated pretext tasks discussed earlier.

Replacing localized aggregation with global self-attention has emerged as a potent strategy for mitigating the over-squashing bottleneck and capturing non-local interactions essential for accurate quantum chemical predictions. Architectures like Graphormer and Edge-augmented Graph Transformers (EGT) demonstrate that global attention can serve as a flexible, adaptive replacement for graph convolution, achieving state-of-the-art performance on large-scale datasets such as PCQM4Mv2 by explicitly encoding pairwise structural information [11; 36]. Furthermore, hybrid architectures that integrate MPNN locality with Transformer globality, such as GPS++, leverage the strengths of both paradigms to optimize expressivity and efficiency in molecular property prediction [54]. These models often incorporate 3D positional encodings and equivariant biases to respect physical symmetries, ensuring that the learned representations remain consistent under roto-translational transformations [55], thus maintaining the geometric fidelity required by the pre-training objectives.

However, the computational cost of quadratic attention scaling poses significant challenges for processing massive scientific graphs. To address this, recent works explore parameter-efficient designs and graph parallelism techniques that distribute input graphs across multiple GPUs, enabling the training of models with hundreds of millions of parameters for atomic simulations [56]. Additionally, the integration of Large Language Models (LLMs) offers a pathway to parameter efficiency and enhanced reasoning. Frameworks like MuseGraph and GraphEdit utilize LLMs for instruction tuning and graph structure learning, allowing for generic graph mining across tasks without extensive retraining [57; 58]. Similarly, compact models like MolE adapt transformer architectures originally designed for language to molecular graphs, employing multi-task pretraining to achieve robust generalization with fewer parameters [59].

Despite these advances, trade-offs remain between architectural expressivity, computational scalability, and data efficiency. While global attention enhances long-range modeling, it often requires sophisticated positional encodings and sparse attention mechanisms to remain tractable [55]. Future directions must focus on developing neuro-symbolic architectures that seamlessly integrate continuous physical fields with discrete graph tokens, fostering unified models capable of zero-shot reasoning across disparate scientific domains [60]. The synthesis of efficient tokenization, hybrid attention mechanisms, and cross-modal integration will define the next generation of scalable GFMs, ultimately accelerating the pace of scientific discovery by enabling robust transfer learning in data-scarce regimes. Building upon these scalable architectures and hybrid attention mechanisms, the transition from specialized graph neural networks to foundation models necessitates robust strategies for adapting pre-trained knowledge to data-scarce scientific domains.

3.3 Cross-Domain Transfer Learning and Task Adaptation Strategies

The transition from specialized graph neural networks to foundation models necessitates robust strategies for adapting pre-trained knowledge to data-scarce scientific domains. While self-supervised pre-training on massive corpora yields universal representations, the efficacy of these models in downstream tasks is fundamentally constrained by distribution shifts between source and target domains, such as the leap from organic molecules to inorganic crystals or from common proteins to rare disease targets. A primary challenge lies in the heterogeneity of graph structures and feature spaces across disciplines. Recent investigations reveal that transfer learning in GNNs is not automatic; success depends critically on the similarity of community structures and the inductive nature of the architecture [61]. To bridge this gap, multi-task learning frameworks have emerged as a pivotal strategy, jointly optimizing diverse auxiliary objectives to learn robust feature extractors that generalize beyond single-domain biases. For instance, structured multi-task approaches leverage task relation graphs to model latent dependencies between properties, significantly enhancing performance in low-data regimes by sharing statistical strength across related scientific endpoints [62]. Furthermore, heterogeneous motif graph neural networks construct explicit connections between molecules and functional subgraphs, enabling effective knowledge transfer even when target datasets are exceedingly small [63].

However, naive fine-tuning often suffers from "catastrophic forgetting" or overfitting to spurious correlations present in limited target data. To mitigate this, recent methodologies focus on learning invariant representations that remain stable under structural shifts. The Cluster Information Transfer (CIT) mechanism, for example, combines cluster-specific and cluster-independent information to enhance generalization across varying graph structures, effectively decoupling domain-specific noise from causal mechanisms [64]. Similarly, causal attention learning strategies disentangle causal features from non-causal shortcuts, ensuring that models rely on mechanistic substructures rather than dataset-specific artifacts, thereby improving out-of-distribution generalization [65]. In scenarios where source and target domains exhibit distinct heterophily levels—common when transferring from homophilic molecular property tasks to heterophilic biological interaction networks—adaptive frameworks like the Heterophily Snowflake Hypothesis allow nodes to determine their own aggregation patterns, facilitating transfer across disparate relational regimes [66].

A critical frontier in task adaptation is addressing size generalization, as pre-trained models often fail when applied to systems significantly larger than those in the training distribution. Regularization techniques such as SizeShiftReg simulate size shifts during training to enforce robustness, while self-supervised tasks designed to learn local structural invariants further improve extrapolation to macroscopic systems [67; 23]. Moreover, the integration of domain adaptation with generative constraints ensures that adapted models respect physical laws; for example, physics-aware multiplex architectures separate local and non-local interactions to maintain accuracy when transferring between small molecules and macromolecular complexes [68]. Looking forward, the development of scalable foundation models like AnyGraph, which utilizes mixture-of-experts architectures to handle feature and structure heterogeneity simultaneously, suggests a path toward zero-shot adaptation across the entire spectrum of scientific discovery [69]. Ultimately, the future of cross-domain transfer lies in synthesizing causal invariant learning with scalable, modular architectures that can dynamically reconfigure their inductive biases to match the physical constraints of novel scientific frontiers.

3.4 Multimodal Integration with Large Language Models and Scientific Literature

Building on the dynamic reconfiguration of inductive biases and the potential for zero-shot adaptation discussed previously, the convergence of Graph Foundation Models (GFMs) with Large Language Models (LLMs) represents the next paradigmatic shift: the evolution from robust structural learners to holistic, multimodal reasoning systems. This integration addresses a critical limitation in traditional graph learning by leveraging the vast semantic knowledge embedded in scientific literature alongside geometric and topological data. By aligning graph structural embeddings with the high-dimensional semantic spaces of LLMs, researchers are developing frameworks capable of cross-modal retrieval, hypothesis generation, and the interpretation of complex scientific protocols.

A primary architectural strategy to achieve this synergy involves cross-modal alignment through projection mechanisms that map graph representations into the token embedding space of pre-trained LLMs. Notable examples include molecular multimodal foundation models that utilize contrastive learning to associate molecular graphs with textual descriptions extracted from scientific citations, enabling tasks such as molecule captioning and property extraction from natural language queries [70]. Similarly, frameworks like MolCA and GIT-Mol employ cross-modal projectors to bridge the modality gap, facilitating zero-shot generalization where textual prompts guide the prediction of molecular properties without task-specific fine-tuning. These approaches often leverage the "graph vocabulary" concept, treating recurring structural motifs as tokens that can be seamlessly processed by language transformers, thereby unifying the representation of discrete scientific entities and continuous textual knowledge [53].

Beyond static alignment, emerging methodologies integrate graph traversal and execution directly into the reasoning loops of LLMs, a paradigm known as Graph-Augmented Reasoning or Chain-of-Thought (CoT). In this context, LLMs act as high-level controllers that iteratively query graph-structured knowledge bases to perform logical deductions. For instance, systems like SciAgents utilize large-scale ontological knowledge graphs interconnected with LLMs to autonomously generate and refine research hypotheses, uncovering hidden interdisciplinary relationships that transcend human-curated boundaries [71]. Furthermore, instruction-tuning frameworks such as MuseGraph distill the reasoning capabilities of LLMs into graph mining tasks, creating dynamic instruction packages that enhance the model's ability to handle diverse graph analytics without retraining [57]. This symbiotic relationship is further evidenced by works demonstrating that LLMs can serve as effective priors for causal graph discovery, reducing the hypothesis space through semantic understanding of variable relationships [72].

The integration also extends to heterogeneous data sources, incorporating visual scientific data such as electron micrographs and spectral plots alongside graph structures and text. Multimodal frameworks like GALLON distill knowledge from LLMs into graph neural networks, synergizing textual and visual insights with structural analysis to improve prediction accuracy and efficiency [73]. However, significant challenges remain, particularly regarding the alignment of disparate inductive biases and the mitigation of hallucinations when LLMs reason over noisy experimental data. While LLMs excel at semantic abstraction, they often struggle with the precise numerical and geometric constraints inherent in scientific graphs, necessitating hybrid architectures that enforce physical consistency [74]. Future directions point toward neuro-symbolic systems where LLMs provide semantic context and hypothesis generation, while graph foundation models ensure structural validity and adherence to physical laws, ultimately accelerating the design-make-test-analyze cycle in autonomous laboratories.

4 Generative Graph Models and Autonomous Discovery Systems

4.1 Architectural Paradigms for Equivariant and Scalable Graph Generation

The architectural landscape for generative graph models in scientific discovery has undergone a paradigmatic shift, moving from sequential, string-based representations to direct, geometry-aware synthesis in three-dimensional Euclidean space. Early approaches, such as autoregressive models that decompose graph generation into node and edge sequences [75], often struggled with the non-uniqueness of graph representations and the inability to inherently capture complex non-local dependencies critical in molecular systems. While reinforcement learning frameworks like the Graph Convolutional Policy Network [7] successfully optimized for specific properties by incorporating domain rules, they frequently relied on 2D topological graphs, neglecting the fundamental 3D geometric constraints that dictate physical stability and chemical reactivity. This limitation has driven the development of equivariant generative architectures that explicitly enforce roto-translational symmetry, ensuring that generated structures remain physically consistent regardless of their spatial orientation.

A pivotal advancement in this domain is the integration of E(n)-equivariance into diffusion and flow-based models. Unlike invariant models that discard directional information, E(n)-Equivariant Graph Neural Networks [24] enable the direct generation of atomic coordinates and features simultaneously, preserving the geometric integrity of the system throughout the denoising process. This capability is further enhanced by steerable architectures [3], which utilize higher-order tensor representations to model complex many-body interactions and angular dependencies that scalar-based message passing fails to resolve. Recent theoretical analyses confirm that such equivariant layers possess superior expressive power in distinguishing geometric graphs compared to their invariant counterparts [5], making them indispensable for generating valid, low-energy conformations in de novo design tasks. Furthermore, the emergence of geometry-complete perceptron networks [76] demonstrates that strictly adhering to SE(3) symmetry yields statistically significant improvements in predicting binding affinities and modeling Newtonian dynamics, underscoring the necessity of geometric priors in generative workflows.

However, the computational cost of enforcing strict equivariance and modeling higher-order interactions poses significant scalability challenges, particularly when addressing large-scale atomistic systems such as proteins or extended crystal lattices. Traditional equivariant models often suffer from memory bottlenecks due to the expansion of feature spaces into high-order tensors. To mitigate this, recent innovations have introduced graph parallelism strategies [56], which distribute input graphs across multiple processing units to train models with billions of parameters. This scaling is complemented by efficient architectural designs, such as local frame methods [77], which project equivariant features into invariant local reference frames, allowing standard GNNs to achieve maximum expressivity with reduced computational overhead. Additionally, the adoption of sparse attention mechanisms and hybrid transformer architectures [55; 11] facilitates the capture of long-range electrostatic interactions without the quadratic scaling associated with global attention, enabling the generation of macroscopic material phases and complex biomolecular assemblies.

Looking forward, the convergence of equivariant generative modeling with scalable foundation model architectures represents the frontier of autonomous discovery. The transition from task-specific generators to unified, pre-trained systems capable of zero-shot generalization across chemical and material domains [9] promises to accelerate the identification of novel catalysts and functional materials. Future research must address the remaining tension between strict physical symmetry enforcement and the flexibility required to model systems with broken symmetries, such as those under external fields [26], while continuing to push the boundaries of model size and data diversity to unlock the full potential of generative AI in scientific inquiry.

4.2 Constraint-Aware Synthesis and Semantic Validity Assurance

While the previous section established the critical role of equivariant architectures in preserving geometric integrity, the generation of novel scientific entities via graph neural networks further necessitates a rigorous adherence to domain-specific logical and chemical constraints. The combinatorial space of possible graphs vastly exceeds the manifold of physically realizable structures, making semantic validity—encompassing chemical valency, geometric stability, and synthesizability—a fundamental architectural requirement rather than a mere post-processing step. Early approaches relying on post-hoc filtering proved computationally wasteful given the low validity rates of unconstrained samples, driving a paradigm shift toward embedding hard constraints directly into the generative process. Grammar-guided and rule-based generation represents a robust strategy for guaranteeing local validity during sequential construction. By integrating molecular hypergraph grammars or context-free grammar constraints with neural decoders, models can enforce valence satisfaction and bond order consistency in real-time. For instance, the Graph Convolutional Policy Network (GCPN) operates within an environment that incorporates domain-specific rules, utilizing reinforcement learning to optimize objectives while strictly obeying physical laws such as chemical valency [7]. Similarly, methods modeling chemical reactions as graph transformations, such as the Graph Transformation Policy Network, extend standard policy gradients with constraints to navigate the discrete space of bond changes effectively [78].

Complementing these discrete approaches, differentiable constraints and penalty regularization offer a continuous optimization pathway to guide generative trajectories toward stable regions of chemical space. Techniques that incorporate physical stability metrics, energy barriers, and synthesizability scores as differentiable loss terms allow for the seamless integration of domain knowledge into the training objective. The GraphDF model exemplifies this by employing a discrete flow model with invertible modulo shift transforms, which eliminates the inaccuracies of dequantization inherent in continuous latent variable models, thereby ensuring the generation of valid discrete graph structures [79]. Furthermore, constrained variational autoencoders shape the latent space to facilitate the design of molecules that are locally optimal in desired properties while maintaining structural coherence [80]. However, a trade-off persists between the rigidity of hard constraints, which may limit exploration, and the flexibility of soft penalties, which risk generating invalid candidates. Hybrid pipelines address this dichotomy by combining unconstrained high-diversity generation with efficient validity correction algorithms. These systems leverage the expressive power of models like those discussed in [81] to propose novel topologies, subsequently applying rapid correction filters to maximize the yield of useful candidates.

Emerging trends indicate a convergence of these constraint-aware methodologies with multi-objective optimization frameworks to address increasingly complex design goals. In the design of high-octane fuels, for example, modular frameworks integrate generative models with Bayesian optimization to navigate continuous molecular spaces while respecting ignition property constraints [82]. Despite these advances, significant challenges remain in scaling constraint-aware synthesis to complex, multi-component systems such as protein-ligand complexes or crystalline materials, where non-local interactions and long-range dependencies complicate validity assurance. This limitation sets the stage for the active search strategies discussed next, where Reinforcement Learning serves as a pivotal paradigm for navigating the combinatorial explosion of inverse design. Future research must focus on developing neuro-symbolic architectures that can dynamically reason over complex logical rules and physical equations during generation, moving beyond static grammars to adaptive, physics-informed synthesis engines capable of discovering entirely new classes of stable matter.

4.3 Reinforcement Learning and Sequential Decision Making for Optimization

Reinforcement learning (RL) has emerged as a pivotal paradigm for navigating the combinatorial explosion inherent in inverse molecular design, framing the generation of graph structures as a sequential decision-making process. Unlike static generative models, RL agents learn policies to iteratively modify graph topologies—adding or removing atoms and bonds—to maximize cumulative rewards defined by target physicochemical properties. The Graph Convolutional Policy Network (GCPN) exemplifies this approach, employing policy gradient methods to optimize domain-specific rewards while adhering to strict chemical valency rules through an environment that enforces physical constraints [7]. By integrating adversarial loss, GCPN ensures generated molecules resemble known chemical scaffolds while achieving significant improvements in property optimization compared to variational autoencoders.

The sequential nature of chemical synthesis and reaction prediction further necessitates RL frameworks capable of modeling complex graph transformations. The Graph Transformation Policy Network (GTPN) addresses chemical reaction product prediction by formulating the process as a sequence of graph edits, learning optimal bond-breaking and bond-forming actions directly from data without assuming fixed transformation orders [78]. This end-to-end learning capability allows the agent to explore non-intuitive reaction pathways that template-based methods might miss. Similarly, in retrosynthesis, RL serves as a powerful search strategy for causal discovery, where agents construct Directed Acyclic Graphs (DAGs) representing reaction sequences by maximizing scores that balance chemical feasibility with strategic efficiency [83].

Beyond discrete graph modifications, RL strategies are increasingly coupled with continuous latent space optimization to refine molecular properties. Hybrid frameworks integrate generative models with Bayesian optimization or genetic algorithms, enabling agents to navigate continuous representations for discovering molecules with extreme properties, such as high octane numbers [82]. This synergy allows for the efficient exploration of vast chemical spaces where direct discrete search is computationally prohibitive. Furthermore, the application of RL extends to bandit optimization problems in drug discovery, where Graph Neural Networks estimate reward functions over graph-structured data with theoretical guarantees on regret bounds, ensuring scalable exploration of large molecular domains [84].

Despite these advancements, significant challenges remain in balancing exploration versus exploitation and ensuring the synthesizability of generated candidates. Standard policy gradient methods often suffer from high variance and mode collapse, potentially limiting the diversity of discovered structures. Recent trends address these limitations by incorporating constrained optimization and multi-objective reward shaping to simultaneously optimize for potency, solubility, and synthetic accessibility. Future directions point toward hierarchical RL agents that operate at multiple scales, from functional group manipulation to global scaffold rearrangement, and the integration of large language models to guide policy search using semantic knowledge from scientific literature. As these systems evolve, the convergence of RL with equivariant graph architectures promises to unlock autonomous discovery pipelines capable of identifying novel materials and therapeutics with unprecedented efficiency and physical consistency.

4.4 Closed-Loop Autonomous Laboratories and AI-Driven Scientific Workflows

Building directly upon the reinforcement learning strategies and surrogate-driven evaluation cycles established previously, closed-loop autonomous laboratories represent the apex of AI-driven scientific discovery. These systems integrate generative graph models, predictive surrogates, and robotic execution into a unified, self-driving workflow, shifting the scientific method from a human-centric, iterative process to a continuous, automated cycle. In this paradigm, hypotheses are generated, experiments are designed and executed, and results are analyzed to refine subsequent iterations without human intervention. The core of this architecture relies on the seamless synergy between the graph-based generative models that propose novel candidates—often utilizing the RL mechanisms discussed earlier—and surrogate models that rapidly evaluate their properties, thereby navigating vast combinatorial spaces with unprecedented efficiency.

Recent advancements have demonstrated the efficacy of coupling these generative graph models with reinforcement learning and Bayesian optimization to drive the "design" phase of the loop. For instance, frameworks utilizing Graph Convolutional Policy Networks (GCPN) have shown significant improvements in optimizing molecular properties while adhering to chemical valency rules, effectively biasing the search toward synthesizable candidates [7]. Similarly, the integration of retrosynthesis predictors, such as RetroGNN, allows these systems to approximate synthetic accessibility in real-time, filtering out chemically unfeasible structures before they reach the robotic synthesis stage [85]. This constraint-aware generation is critical for minimizing the "make-test" cycle time in physical laboratories, ensuring that the theoretical optimizations derived from RL agents translate into physically realizable experiments.

The "test" and "analyze" phases are increasingly empowered by graph neural network (GNN) surrogates that replace computationally expensive physics-based simulations, closing the feedback loop with high-speed inference. Models like MeshGraphNets and GNN-Surrogate have proven capable of learning mesh-based simulations and predicting complex physical dynamics orders of magnitude faster than traditional solvers, enabling real-time feedback for experimental parameter adjustment [42; 43]. Furthermore, the incorporation of probabilistic frameworks such as GFlowNets facilitates the efficient exploration of hypothesis spaces by sampling diverse, high-reward candidates based on unnormalized probability distributions, addressing the challenge of epistemic uncertainty in experimental design [86]. These generative flows ensure that the autonomous agent does not merely exploit known high-performing regions but actively explores novel chemical or material spaces, balancing the exploration-exploitation trade-off inherent in the RL policies discussed prior.

A transformative trend in this domain is the integration of Large Language Models (LLMs) as cognitive agents that orchestrate the entire workflow, adding a layer of semantic reasoning to the structural capabilities of GNNs. Systems like SciAgents leverage ontological knowledge graphs and multi-agent reasoning to autonomously generate research hypotheses, critique existing designs, and translate abstract scientific goals into executable graph-based parameters [71]. This multimodal approach bridges the gap between unstructured scientific literature and structured graph data, allowing agents to formulate hypotheses grounded in broad domain knowledge [70]. The vision of "Level 5" autonomy, where AI systems achieve Nobel-quality discoveries with zero human intervention, relies heavily on such synergistic architectures that combine the structural reasoning of GNNs with the semantic flexibility of LLMs [87].

Despite these advances, significant challenges remain in ensuring the robustness and generalizability of closed-loop systems. The reliability of surrogate models in out-of-distribution regimes and the physical consistency of generated candidates are critical bottlenecks that must be addressed to fully realize the potential of these pipelines. Future directions must focus on developing more rigorous uncertainty quantification methods within the loop and enhancing the interpretability of agent decisions to build trust in fully autonomous discovery pipelines. As these systems evolve, they promise to democratize scientific inquiry, accelerating the pace of discovery in fields ranging from drug development to materials science by systematically exploring the frontiers of the unknown.

5 Trustworthiness, Interpretability, and Causal Reasoning in Scientific Applications

5.1 Mechanistic Interpretability and Subgraph Attribution

Mechanistic interpretability in scientific graph learning transcends mere feature attribution, demanding the isolation of causal structural motifs—such as functional groups in chemistry or protein domains in biology—that physically drive model predictions. The field has bifurcated into post-hoc explanation frameworks, which analyze trained black-box models, and intrinsic architectures designed for self-explainability. Post-hoc methods predominantly rely on optimizing subgraph masks to maximize mutual information between the explanation and the prediction. Seminal work like [88] formalizes this by identifying compact subgraphs and node features crucial for prediction, while [89] employs Hilbert-Schmidt Independence Criterion (HSIC) Lasso to learn nonlinear interpretable models locally. However, these edge- or node-level attributions often fail to capture higher-order interactions essential in scientific systems. To address this, [90] introduces SubgraphX, utilizing Monte Carlo tree search and Shapley values to explicitly identify important subgraphs, capturing coalition effects that linear attributions miss. Similarly, [91] extracts relevant walks via nested attribution schemes to reveal complex structure-property relationships in quantum chemistry.

Despite their utility, post-hoc methods suffer from a fundamental disconnect: the explanation is not part of the model's reasoning process, potentially leading to unfaithful rationales. Consequently, recent paradigms shift toward intrinsic interpretability, where explanations are derived directly from the model's decision logic. [92] integrates prototype learning, classifying inputs based on similarity to learned latent prototypes that correspond to recognizable chemical motifs. Complementing this, [93] introduces differentiable concept-discovery modules that force the network to solve tasks using explicitly discovered graph concepts, thereby supporting effective interventions. Furthermore, [94] extends Generalized Additive Models to graphs, offering fully transparent visualizations of feature relationships without sacrificing accuracy. A critical advancement in this domain is the transition from associative to causal explanations. Standard attribution often confounds spurious correlations with causal mechanisms, limiting out-of-distribution generalization. [95] frames explanation as a causal learning task using Granger causality, while [96] employs reinforcement learning to sequentially construct explanatory subgraphs that account for edge dependencies and coalition effects.

The evaluation of these mechanisms remains a significant challenge, particularly in scientific domains where ground-truth explanatory subgraphs are rarely available. [97] proposes a holistic benchmark encompassing faithfulness, sparsity, and plausibility across diverse molecular and protein datasets. Yet, standard fidelity metrics often ignore the distributional shift between the full graph and the explanatory subgraph. [98] addresses this by employing front-door adjustment to deconfound the evaluation process, ensuring that identified substructures are causally relevant rather than statistical artifacts. Looking forward, the integration of domain-specific constraints into explanation generation is paramount. Methods like [99] and [100] demonstrate that enforcing chemical validity at the motif level yields explanations that are not only faithful to the model but also scientifically actionable. Future research must further bridge the gap between neural attribution and symbolic physical laws, ensuring that identified subgraphs align with established mechanistic theories in physics and biology.

5.2 Causal Discovery and Invariant Learning for Robust Generalization

Building on the discussion of mechanistic interpretability, which isolates causal structural motifs through intrinsic architectures and domain constraints, the field must now address the fundamental challenge of distinguishing these invariant mechanisms from spurious correlations to ensure robust scientific discovery. The transition from associative pattern recognition to causal mechanism identification represents a pivotal evolution in applying Graph Neural Networks (GNNs). Standard GNN architectures, while effective under independent and identically distributed (I.I.D.) assumptions, frequently exploit spurious correlations inherent in biased scientific datasets, leading to catastrophic failures when encountering out-of-distribution (OOD) experimental conditions [101]. To ensure robust generalization, recent paradigms integrate causal inference principles to disentangle invariant causal rationales from environment-specific confounders. A primary strategy involves Invariant Rationale Learning, where models are trained to identify subgraphs that maintain predictive stability across multiple interventional distributions. For instance, the DIR-GNN framework explicitly intervenes on training data to isolate causal substructures, filtering out unstable shortcut features that correlate with labels only within specific domains [102]. Similarly, approaches like Causal Attention Learning (CAL) parameterize backdoor adjustments to mitigate the confounding effects of non-causal features, thereby enforcing a stable relationship between causal patterns and predictions regardless of distribution shifts [65].

Beyond feature disentanglement, the direct recovery of causal structures from high-dimensional observational data remains a critical challenge. Traditional constraint-based methods often struggle with scalability and non-linear dynamics common in complex biological and physical systems. Amortized causal discovery leverages the expressive power of GNNs to overcome these limitations. Methods such as DAG-GNN employ variational autoencoders with structural constraints to learn Directed Acyclic Graphs (DAGs) from non-linear data, significantly outperforming linear structural equation models [103]. Furthermore, emerging foundation model perspectives propose pre-training deep learners to resolve predictions from classical algorithms on variable subsets, achieving orders-of-magnitude speedups while generalizing to unseen data-generating mechanisms [104]. The integration of Large Language Models (LLMs) further enhances this process by providing informative priors on edge directionality, effectively reducing the hypothesis space for causal graph discovery [72].

Addressing hidden confounders, which violate the strong ignorability assumption, requires innovative instrumental variable (IV) strategies. The CgNN framework uniquely utilizes network topology itself as an instrumental variable, combining GNNs with attention mechanisms to mitigate bias from unobserved confounders in networked data [105]. This is particularly vital in domains like genomics and epidemiology, where unmeasured latent factors frequently distort treatment effect estimates. Additionally, frameworks like REASON utilize hierarchical GNNs to model both intra-level and inter-level causal dependencies, enabling precise root cause localization in complex multi-scale systems [106].

Despite these advancements, significant challenges persist. The identifiability of causal graphs from observational data alone remains theoretically limited without strong assumptions or interventional data [107]. Moreover, many current methods assume specific noise models or functional forms that may not hold in all scientific contexts. Future research must focus on developing neuro-symbolic hybrids that embed physical laws as hard causal constraints, ensuring that learned graphs are not only statistically robust but also mechanistically consistent with known scientific principles. Crucially, establishing these invariant causal foundations enables the operationalization of counterfactual reasoning discussed in the following section. By synthesizing counterfactual reasoning with invariant learning, the field moves beyond static identification toward active hypothesis generation, bridging the gap between data-driven correlation and true scientific causation to enable AI systems that generate reliable hypotheses even in data-scarce, high-stakes environments [108; 47].

5.3 Counterfactual Reasoning and Hypothesis Validation

Counterfactual reasoning in scientific graph learning transcends traditional post-hoc interpretability by actively probing the causal mechanisms underpinning model predictions through minimal structural interventions. Unlike associative methods that merely highlight salient features, counterfactual approaches seek to answer "what-if" questions: identifying the smallest modification to a molecular graph or physical system that would alter a predicted property, thereby validating the robustness of learned representations and distinguishing causal drivers from spurious correlations. Formally, given a graph G and a predictor f, a counterfactual explanation seeks a perturbed graph G' such that $f(G) \neq f(G')$ while minimizing a distance metric $d(G, G')$, often constrained to preserve chemical validity or physical laws.

Recent advancements have formalized this optimization to ensure explanations are both minimal and faithful. [109] introduced a framework generating counterfactuals via edge deletions, demonstrating that removing fewer than three edges on average is often sufficient to flip predictions in benchmark datasets, thus isolating critical structural determinants. However, purely topological perturbations risk generating chemically invalid structures. To address this, [49] and related generative strategies emphasize motif-consistent modifications, ensuring counterfactuals remain on the data manifold and adhere to valency rules. This shift from arbitrary edge flipping to semantic motif manipulation is crucial for scientific utility, as it yields hypotheses that are experimentally actionable rather than merely mathematically optimal.

The integration of counterfactuals into hypothesis validation extends beyond explanation to active learning and causal discovery. [110] proposed a unified framework (CF^2) that leverages both factual and counterfactual reasoning to evaluate the necessity and sufficiency of explanations without ground-truth labels, providing a rigorous metric for model trustworthiness in data-scarce regimes. Furthermore, counterfactual data augmentation has emerged as a powerful technique for enhancing model robustness. By synthesizing counterfactual links and structures during training, models can explicitly learn invariant causal relationships, mitigating the impact of distribution shifts common in scientific datasets [108]. This approach aligns with causal inference principles, where interventions are used to disentangle confounding variables, as seen in frameworks that synthesize unbiased training samples to decorrelate causal and bias substructures [111].

Despite these advances, significant challenges remain in scaling counterfactual reasoning to complex, continuous physical systems and ensuring the semantic realism of generated scenarios. Current methods often struggle with the combinatorial explosion of valid perturbations in large biomolecules or materials. Future directions must integrate neuro-symbolic constraints to guarantee that counterfactual trajectories respect conservation laws and thermodynamic stability. Moreover, the synergy between counterfactual generation and autonomous laboratory workflows offers a transformative potential: by simulating minimal structural modifications in silico, AI agents can prioritize high-value wet-lab experiments, effectively closing the loop between computational hypothesis generation and empirical validation. As the field evolves, the fusion of counterfactual logic with large language models and knowledge graphs will likely enable more sophisticated scientific reasoning, allowing models to not only predict outcomes but to articulate the causal narratives driving scientific discovery [71].

5.4 Uncertainty Quantification and Robustness in Noisy Regimes

Bridging the gap between counterfactual hypothesis generation and neuro-symbolic constraint enforcement, the reliability of Graph Neural Networks (GNNs) in scientific discovery ultimately hinges on rigorous uncertainty quantification (UQ) and robustness. While counterfactual reasoning identifies minimal structural interventions to validate causal mechanisms, and neuro-symbolic methods embed physical laws to ensure plausibility, both paradigms require calibrated confidence bounds to guide high-stakes decisions in data-scarce, noisy, and distributionally shifted regimes. Whether identifying viable drug candidates or predicting material stability, scientific applications demand a clear distinction between aleatoric uncertainty, arising from inherent data noise, and epistemic uncertainty, stemming from model limitations. Unified Bayesian frameworks have been proposed to aggregate these sources, providing a total predictive uncertainty metric essential for trustworthy deployment [112].

Bayesian approaches offer a principled foundation for this quantification by treating model parameters or graph structures as random variables. For instance, Graph Posterior Networks (GPN) explicitly perform Bayesian posterior updates for interdependent nodes, adhering to axioms that characterize expected uncertainty behavior in homophilic graphs while effectively detecting anomalous features and out-of-distribution (OOD) classes [113]. Complementing this, Bayesian GNNs with adaptive connection sampling generalize stochastic regularization, jointly learning sampling rates to alleviate over-smoothing and provide robust uncertainty estimates without manual hyperparameter tuning [114]. Furthermore, interpretable Bayesian neural networks have been adapted for graph structure learning, enabling high-fidelity posterior approximation via Markov Chain Monte Carlo to quantify uncertainty in edge predictions, a critical capability when the underlying relational structure is unknown [115].

Beyond purely Bayesian methods, frequentist and ensemble-based strategies provide computationally efficient alternatives for scaling UQ to larger scientific datasets. Deep ensembles and committee-based methods, such as those utilizing jackknife estimators, derive deterministic confidence intervals by measuring prediction variance across multiple model instances. This approach is particularly effective in mitigating the sensitivity of models to noisy graph constructions, as demonstrated in bootstrapped G-CNNs that reduce reliance on specific graph topologies in biomedical applications [116]. Additionally, energy-based OOD detection frameworks like GNNSafe leverage energy functions extracted from standard classifiers to distinguish in-distribution samples from OOD data with provable margins, addressing the risk of overconfident predictions on shifted distributions [117]. Another approach involves multi-source uncertainty frameworks that utilize Graph-based Kernel Dirichlet distribution Estimation (GKDE) to detect misclassifications and OOD nodes by modeling various types of predictive uncertainties [118].

Robustness in these noisy regimes also necessitates architectural innovations that explicitly account for data imperfections, ensuring that uncertainty estimates remain meaningful even under adversarial conditions. Methods like NOSMOG integrate adversarial feature augmentation and representational similarity distillation to train noise-robust, structure-aware MLPs, overcoming the scalability and noise sensitivity limitations of traditional GNNs [119]. Similarly, spectral approaches such as EvenNet improve robustness against structural attacks and homophily variations by employing even-polynomial graph filters that ignore odd-hop neighbors, thereby stabilizing performance across diverse graph regimes [120]. For dynamic systems, probabilistic forecasting models like Graph-EFM utilize hierarchical graph constructions and latent variables to capture spatially coherent uncertainty in chaotic environments, ensuring that ensemble forecasts accurately reflect predictive confidence [121].

Despite these advances, significant challenges remain in scaling UQ methods to massive molecular datasets and seamlessly integrating them into autonomous discovery loops. The synergy with the preceding discussion on counterfactuals suggests a need to disentangle spurious correlations from true mechanistic uncertainty, while the transition to neuro-symbolic frameworks requires uncertainty estimates that respect physical constraints. Future research must prioritize the development of lightweight UQ surrogates that maintain calibration under extreme distribution shifts and the integration of causal inference to ensure that generative models like GFlowNets can efficiently explore vast chemical spaces with reliable error bars [86; 122]. By addressing these gaps, the field can move toward unified frameworks where uncertainty quantification, causal reasoning, and physical constraints coalesce to enable truly autonomous and trustworthy scientific agents.

5.5 Neuro-Symbolic Integration and Physics-Informed Constraints

The convergence of neural representation learning with symbolic reasoning and physical laws represents a critical frontier for enhancing the trustworthiness of Graph Neural Networks (GNNs) in scientific discovery. While purely data-driven models excel at pattern recognition, they often fail to guarantee adherence to fundamental conservation laws or logical consistency, limiting their utility in high-stakes scientific applications. Neuro-symbolic integration addresses this by embedding hard constraints, differential operators, and logical rules directly into the graph learning architecture, ensuring predictions are not only accurate but also physically plausible and interpretable.

A primary paradigm involves replacing direct state prediction with learned constraint functions that are solved via optimization, effectively turning the GNN into a differentiable physics engine. For instance, frameworks utilizing graph exterior calculus construct differential operators to embed partial differential equations (PDEs) directly into the message-passing mechanism, enabling the solution of forward and inverse problems on unstructured meshes while respecting underlying physics [44]. Similarly, approaches like Lagrangian Graph Neural Networks learn the system's Lagrangian directly from trajectory data, decoupling kinetic and potential energies to inherently satisfy conservation of energy and momentum without explicit penalty terms [41]. Comparative analyses indicate that such architectures exhibit superior zero-shot generalization to system sizes an order of magnitude larger than training data, outperforming unconstrained baselines in long-term rollout stability [39].

Beyond continuous physical laws, neuro-symbolic methods integrate discrete logical reasoning to enhance interpretability. By modeling GNN inference steps as logic programs or utilizing Markov Logic Networks, researchers can verify reasoning paths and incorporate prior domain knowledge as soft or hard constraints. The ExpressGNN framework, for example, leverages GNNs for efficient variational inference in Markov Logic Networks, balancing the representation power of neural networks with the logical rigor of probabilistic logic reasoning [123]. Furthermore, activation rule mining in hidden layers allows for the redescription of neural features into interpretable logical patterns, bridging the gap between sub-symbolic learning and human-understandable scientific concepts [124].

A transformative application of this integration is the distillation of implicit neural representations into explicit symbolic laws. By encouraging sparse latent representations during training, symbolic regression can be applied to extract analytic equations from trained GNNs, effectively rediscovering known physical laws such as Newton's law of gravitation or identifying novel constitutive relations in complex materials [47; 46]. Recent advancements in Kolmogorov-Arnold Networks (KANs) further facilitate this synergy, offering architectures capable of revealing modular structures and compiling symbolic formulas directly into neural weights, thus enabling bidirectional flow between scientific theory and data-driven learning [125].

Despite these advances, challenges remain in balancing computational efficiency with the rigor of constraint enforcement. Strict equivariance or hard logical constraints can restrict the model's expressive capacity or introduce optimization difficulties, particularly in systems with broken symmetries or noisy data. Emerging trends suggest a shift toward "soft" neuro-symbolic frameworks, such as subequivariant networks that relax continuous symmetry constraints to accommodate external fields like gravity, or adaptive rewiring strategies that modify graph topology based on particle system analogies to prevent over-smoothing while maintaining physical fidelity [25; 126]. Future research must focus on developing unified frameworks that seamlessly integrate continuous physical fields, discrete logical rules, and causal mechanisms to create truly autonomous scientific reasoning agents.

6 Domain-Specific Applications and Evaluation Frameworks

6.1 Molecular Design, Cheminformatics, and Pharmaceutical Discovery

The application of Graph Neural Networks (GNNs) in molecular design and cheminformatics represents a paradigm shift from heuristic descriptor-based modeling to end-to-end learning of structural representations. Historically, molecular "fingerprints" served as the standard for encoding structural information, yet these fixed representations inevitably emphasize specific architectural aspects while obscuring others, limiting the model's capacity for data-driven feature discovery [127]. In contrast, modern GNN architectures operate directly on the molecular graph, leveraging message-passing mechanisms to capture complex topological and geometric dependencies. Empirical benchmarks across diverse industrial and public datasets indicate that learned graph convolutions consistently match or exceed the performance of expert-crafted descriptors, particularly in generalizing to novel chemical spaces [128]. However, the transition to foundation models has revealed that self-supervised pretraining benefits are contingent upon dataset scale and architectural complexity, with significant gains observed primarily when scaling to billions of parameters or utilizing massive unlabeled corpora [129; 9].

A critical frontier in pharmaceutical discovery is the prediction of protein-ligand interactions, a bi-level problem requiring the integration of small-molecule graphs with macromolecular structures. Traditional docking simulations are computationally prohibitive for high-throughput screening, prompting the development of deep surrogate models. Architectures such as [130] utilize GNNs to approximate docking scores with orders-of-magnitude speedup, while [131] demonstrates that simplified graph formulations can achieve state-of-the-art accuracy by focusing on non-covalent interaction networks. Furthermore, equivariant models like [28] introduce virtual nodes to explicitly model binding pockets, enhancing the detection of geometric features essential for accurate affinity prediction. Despite these advances, reliability under distributional shift remains a paramount concern; models often exhibit overconfident mis-predictions when encountering out-of-distribution scaffolds, necessitating distance-aware architectures like GNN-SNGP to ensure calibrated uncertainty estimates [132].

Beyond prediction, the inverse design of novel therapeutics via generative models has gained traction. Early approaches utilized Variational Autoencoders and Reinforcement Learning (RL) to navigate chemical space, with frameworks like [7] optimizing for drug-likeness and synthetic accessibility through policy gradients. More recent innovations employ discrete flow models, such as [79], which mitigate the inaccuracies of continuous latent spaces by modeling graph generation as a sequence of discrete transformations. These generative systems are increasingly integrated into closed-loop optimization pipelines, where Bayesian optimization guides the exploration of latent spaces to identify molecules with extreme properties, such as high octane numbers [82]. Nevertheless, challenges persist in ensuring the physical validity and synthesizability of generated candidates. While grammar-guided and constraint-aware methods improve local validity, the field must address the "reality gap" between in silico optimization and experimental feasibility. Future directions point toward multimodal foundation models that align graph structures with textual scientific knowledge [70] and neuro-symbolic frameworks that embed causal reasoning to distinguish true structure-activity relationships from spurious correlations, thereby accelerating the translation of computational designs into clinical candidates.

6.2 Materials Science, Condensed Matter Physics, and Energy Systems

Bridging the methodological foundations established in organic molecular design, the application of Graph Neural Networks (GNNs) to materials science and condensed matter physics introduces distinct geometric complexities required for discovering functional materials and energy systems. Unlike the discrete molecular graphs of cheminformatics, solid-state materials necessitate the explicit encoding of periodic boundary conditions, crystal symmetries, and long-range many-body interactions. Recent advancements have demonstrated that GNNs can approximate Density Functional Theory (DFT) calculations with near-ab initio accuracy while reducing computational costs by orders of magnitude. For instance, scalable architectures like [133] have overcome the over-smoothing limitations of traditional message-passing networks, enabling the training of very deep models (up to 30 layers) that achieve state-of-the-art performance in predicting band gaps and formation energies across diverse energy datasets. Furthermore, the integration of angular information, often neglected in pairwise models, has proven critical; the [134] framework introduces a dual-scale neighbor partitioning mechanism that efficiently captures both bond distances and angles, significantly improving inference speed and accuracy for crystal property prediction.

A central challenge in this domain, mirroring the distributional shift concerns in drug discovery, is the generalization to out-of-distribution (OOD) materials where test samples deviate structurally from the training distribution. Benchmark studies reveal that standard random-split evaluations often yield artificially inflated performance metrics, masking a significant generalization gap. Research indicates that while models like CGCNN and ALIGNN exhibit robustness in OOD settings due to their physical inductive biases, many state-of-the-art architectures fail when extrapolating to novel chemical spaces [135]. To address this, the field is increasingly adopting unified representations such as the [136], which bridges the divide between stoichiometry-only and structure-based descriptors, facilitating transfer learning across representation domains. Moreover, the advent of large-scale datasets like [137] has pushed the boundaries of graph ML towards inorganic nanomaterials, introducing complexities related to graph sizes ranging from tens to hundreds of thousands of nodes, thereby necessitating novel scalable training strategies.

Beyond static property prediction, geometric GNNs are revolutionizing the simulation of atomic dynamics through Machine-Learned Force Fields (MLFFs), providing the dynamic counterpart to the static affinity predictions seen in protein-ligand interactions. By enforcing SE(3) equivariance, these models ensure that predicted forces and energies respect fundamental physical symmetries, a prerequisite for stable molecular dynamics simulations. The [138] model, developed on the massive Open Catalyst 2020 dataset, exemplifies this progress, outperforming previous baselines by 16% while reducing training time by a factor of ten through optimized message-passing schemes that capture complex catalyst-adsorbate interactions. Similarly, [27] challenges the conventional wisdom of hard-coded physical constraints, demonstrating that expressive models trained with physics-based data augmentation can implicitly learn rotational covariance, offering a flexible alternative for large-scale quantum calculations. The scaling of these models to billion-parameter regimes via graph parallelism, as explored in [56], has further unlocked the potential to model higher-order interactions (triplets and quadruplets) essential for accurate catalytic turnover predictions.

Looking forward, the integration of generative capabilities with these predictive surrogates promises fully autonomous materials discovery loops, extending the inverse design paradigms from organic molecules to the combinatorial complexity of inorganic lattices and space groups. The emergence of multimodal agents capable of reasoning over ontological knowledge graphs, as seen in [71], suggests a future where AI systems not only predict properties but also hypothesize novel material compositions by uncovering hidden interdisciplinary relationships. However, realizing this vision demands rigorous evaluation frameworks that prioritize physical consistency and experimental synthesizability over mere predictive accuracy. This emphasis on robustness and causal validity sets the stage for the parallel challenges found in biological systems, where extreme heterogeneity and multi-scale hierarchies require similarly rigorous approaches to ensure scientific reliability.

6.3 Biological Networks, Genomics, and Precision Medicine

The application of graph neural networks (GNNs) to biological networks and genomics represents a paradigm shift from reductionist statistical methods to holistic, system-level modeling of cellular machinery. Unlike chemical graphs with fixed valency rules, biological graphs are characterized by extreme heterogeneity, dynamic topology, and multi-scale hierarchies spanning from nucleotide sequences to tissue-level interactomes. A primary challenge lies in integrating multi-omics data—transcriptomics, proteomics, and epigenomics—into unified representations. Recent frameworks address this by constructing heterogeneous multi-layer graphs that capture both intra-omic correlations and inter-omic regulatory dependencies. For instance, [139] demonstrates that leveraging both inter- and intra-omic connections within a single GNN architecture significantly outperforms early or late fusion strategies in cancer subtype classification, highlighting the necessity of preserving topological integrity across data modalities. Similarly, [140] introduces a multimodal similarity learning approach that unifies gene representations from single-cell sequencing and spatial transcriptomics, effectively capturing functional similarities across diverse tissues and species despite data sparsity.

In the realm of disease mechanism elucidation, particularly cancer genomics, the ability to model synthetic lethality and prioritize driver genes is critical. Traditional methods often rely on single-network analyses, which can yield conflicting predictions due to network noise or incompleteness.

To mitigate this, [141] proposes an explainable multilayer GNN that aggregates information from multiple gene-gene interaction networks, achieving superior precision in identifying cancer genes while providing mechanistic insights through gene set enrichment. Furthermore, the prediction of protein-protein interactions (PPIs) remains fundamental for mapping disease pathways. While standard GNNs struggle with generalization to novel proteins, [142] introduces a correlation-based GNN framework that explicitly models inter-novel-protein interactions, addressing the distribution shift inherent in discovering unknown biological relationships. The complexity of biological systems also necessitates modeling higher-order interactions beyond pairwise edges. [143] utilizes hypergraph neural networks to represent genetic pathways as hyperedges, enabling the capture of multi-gene functional modules and improving inductive subgraph prediction for patient-specific mutation profiles.

Precision medicine further demands the integration of spatial context and clinical phenotypes. Spatial multi-modal omics technologies have spurred the development of dynamic graph models like [144], which constructs adaptive graphs to resolve latent semantic relations between sequencing spots, effectively denoising perturbations inherent in biological assays. In clinical diagnostics, graph models are increasingly applied to whole-slide images and patient similarity networks. [145] exemplifies this by predicting HER2 status directly from whole-slide histology images using graph representations of tissue regions, offering a non-invasive alternative to expensive genomic testing. Additionally, [146] argues against fixed patient similarity metrics, proposing an end-to-end learnable graph structure that dynamically optimizes patient connections for diagnostic accuracy. Despite these advances, challenges persist in handling the "small n, large p" nature of genomic data and ensuring causal interpretability. Future directions must move beyond associative learning toward causal graph discovery, as initiated by [147], to distinguish true regulatory drivers from confounding correlations, thereby enabling robust, personalized therapeutic strategies.

6.4 Benchmarking Protocols, Reproducibility, and Evaluation Standards

Building on the necessity for causal interpretability and biological consistency highlighted in biological applications, the rigorous evaluation of Graph Neural Networks (GNNs) in scientific discovery demands a paradigm shift from standard predictive accuracy metrics to frameworks that assess physical consistency, generalization capability, and experimental viability. While benchmarks like the Open Graph Benchmark (OGB) and TUDataset have standardized comparisons for generic graph tasks, they often fail to capture the unique constraints of scientific domains, such as geometric equivariance and conservation laws [148; 149]. A critical pitfall in current evaluation protocols is the reliance on random data splits, which frequently results in artificially inflated performance due to dataset redundancy and distributional overlap. Recent studies in materials science demonstrate that models achieving state-of-the-art results under random splitting significantly underperform when evaluated on out-of-distribution (OOD) splits designed to test extrapolation to novel chemical spaces or crystal structures [135]. Consequently, the community must adopt OOD benchmarking suites, such as DrugOOD, which explicitly incorporate noise annotations and domain shifts to reflect the realities of experimental data curation in drug discovery [150].

Reproducibility remains a formidable challenge within this landscape, exacerbated by inconsistent hyperparameter tuning and training procedures that obscure the true contribution of architectural innovations. Empirical re-evaluations reveal that simpler GNN architectures often match or exceed sophisticated counterparts when training protocols are standardized, suggesting that reported gains are frequently artifacts of improper baselines rather than genuine methodological advances [151; 152]. This issue is particularly acute in heterogeneous graph learning, where divergent data processing pipelines hinder fair comparison, prompting the creation of unified benchmarks like HGB to enforce standardized splits and feature processing [153]. Furthermore, the evaluation of explainability methods lacks systematic rigor, often relying on synthetic datasets that do not reflect the complexity of biological or physical systems. New frameworks propose quantitative metrics based on domain-specific concepts, such as pathological measurability, to ensure that explanations align with scientific intuition rather than statistical artifacts [154; 155].

Beyond predictive fidelity, evaluation standards must integrate physical laws as first-class metrics to ensure scientific validity. In dynamical systems, models should be assessed not only on rollout error but also on their adherence to conserved quantities like energy and momentum, as well as their zero-shot generalization to system sizes exceeding training distributions [39; 25]. For generative tasks, success metrics must extend beyond validity scores to include synthesizability and experimental success rates within closed-loop autonomous laboratories. The emergence of virtual environments like DISCOVERYWORLD offers a promising avenue for evaluating end-to-end scientific reasoning agents, measuring their ability to formulate hypotheses and design experiments rather than merely predicting properties [156]. Future benchmarking efforts must therefore converge on multi-dimensional evaluation protocols that jointly optimize for accuracy, physical plausibility, robustness to distribution shifts, and computational efficiency, ensuring that machine learning progress translates into tangible scientific acceleration [138; 9].

7 Conclusion

This subsection synthesizes the transformative trajectory of Graph Neural Networks (GNNs) and Graph Foundation Models (GFMs) in scientific discovery, transitioning from task-specific architectures to unified, generalizable paradigms. The evolution from shallow graph kernels to deep equivariant models has fundamentally altered the landscape of computational science, enabling the precise modeling of physical symmetries and complex many-body interactions inherent in molecular and material systems. As demonstrated by [157], the integration of relational inductive biases allows these models to achieve combinatorial generalization, a hallmark of human-like scientific reasoning that surpasses the limitations of standard Euclidean deep learning. Concurrently, the emergence of geometric GNNs has addressed the critical need for E(3) equivariance, ensuring that predictions regarding atomic forces and crystal properties remain consistent under rotational and translational transformations, as rigorously characterized by the geometric Weisfeiler-Leman test [5].

Despite these architectural triumphs, significant gaps persist in scaling and data efficiency. While recent studies indicate that scaling GNNs to billion-parameter regimes yields substantial performance gains in molecular property prediction [9], the benefits of self-supervised pre-training remain contingent on dataset diversity and objective design, with some evidence suggesting diminishing returns in low-data regimes without careful hyperparameter optimization [129]. The path forward necessitates a shift toward true foundation models capable of zero-shot transfer across disparate domains. Innovations such as unified graph tokenization and scalable graph transformers are pivotal in this regard, offering mechanisms to encode invariant structural patterns that facilitate cross-domain adaptability [8; 53]. Furthermore, the integration of multimodal data, particularly aligning graph structures with textual scientific knowledge via Large Language Models, promises to bridge the gap between structural representation and semantic reasoning, enabling systems to hypothesize novel materials by synthesizing information from vast literature corpora [70].

A critical frontier lies in the convergence of neural learning with symbolic reasoning and causal inference. Current models often rely on spurious correlations; thus, future architectures must prioritize invariant rationale learning to dissect causal mechanisms from confounding variables [102]. The synergy between neuro-symbolic approaches and physics-informed constraints offers a robust pathway toward discovering interpretable physical laws directly from data, moving beyond black-box prediction to formal theorem proving and equation discovery [47; 125]. Moreover, the deployment of generative frameworks like GFlowNets and autonomous agents represents a paradigm shift from passive prediction to active experimental design, optimizing the exploration of vast chemical spaces under uncertainty [86; 71]. Ultimately, realizing the full potential of AI in science requires not only architectural innovation but also a commitment to trustworthiness, ensuring that automated discovery systems are robust, fair, and aligned with the rigorous standards of the scientific method [158].

References

[1] Gated Graph Sequence Neural Networks

[2] Towards Graph Foundation Models A Survey and Beyond

[3] Geometric and Physical Quantities Improve E(3) Equivariant Message Passing

[4] A Hitchhiker's Guide to Geometric GNNs for 3D Atomic Systems

[5] On the Expressive Power of Geometric Graph Neural Networks

[6] Boosting the Cycle Counting Power of Graph Neural Networks with I^2-GNNs

[7] Graph Convolutional Policy Network for Goal-Directed Molecular Graph Generation

[8] OpenGraph Towards Open Graph Foundation Models

[9] On the Scalability of GNNs for Molecular Graphs

[10] A Survey of Graph Meets Large Language Model Progress and Future Directions

[11] Benchmarking Graphormer on Large-Scale Molecular Modeling Datasets

[12] On the Connection Between MPNN and Graph Transformer

[13] Graph neural networks for materials science and chemistry

[14] Few-Shot Graph Learning for Molecular Property Prediction

[15] How Powerful are Graph Neural Networks

[16] Demystifying Higher-Order Graph Neural Networks

[17] Graph Neural Networks Exponentially Lose Expressive Power for Node Classification

[18] Improving Graph Neural Network Expressivity via Subgraph Isomorphism Counting

[19] Relational Pooling for Graph Representations

[20] Provably Powerful Graph Networks

[21] Weisfeiler and Leman go sparse Towards scalable higher-order graph embeddings

[22] Boosting Graph Structure Learning with Dummy Nodes

[23] From Local Structures to Size Generalization in Graph Neural Networks

[24] E(n) Equivariant Graph Neural Networks

[25] Learning Physical Dynamics with Subequivariant Graph Neural Networks

[26] Relaxing Continuous Constraints of Equivariant Graph Neural Networks for Physical Dynamics Learning

[27] ForceNet A Graph Neural Network for Large-Scale Quantum Calculations

[28] VN-EGNN E(3)-Equivariant Graph Neural Networks with Virtual Nodes Enhance Protein Binding Site Identification

[29] Hyper-SAGNN a self-attention based graph neural network for hypergraphs

[30] HyperGCN A New Method of Training Graph Convolutional Networks on Hypergraphs

[31] Hypergraph Node Classification With Graph Neural Networks

[32] Can Graph Neural Networks Count Substructures

[33] Hierarchical Graph Pooling with Structure Learning

[34] Towards Sparse Hierarchical Graph Classifiers

[35] Memory-Based Graph Networks

[36] Global Self-Attention as a Replacement for Graph Convolution

[37] Shortest Path Networks for Graph Property Prediction

[38] How Universal Polynomial Bases Enhance Spectral Graph Neural Networks: Heterophily, Over-smoothing, and Over-squashing

[39] Unravelling the Performance of Physics-informed Graph Neural Networks for Dynamical Systems

[40] SEGNO Generalizing Equivariant Graph Neural Networks with Physical Inductive Biases

[41] Learning the Dynamics of Particle-based Systems with Lagrangian Graph Neural Networks

[42] Learning Mesh-Based Simulation with Graph Networks

[43] GNN-Surrogate A Hierarchical and Adaptive Graph Neural Network for Parameter Space Exploration of Unstructured-Mesh Ocean Simulations

[44] Scalable algorithms for physics-informed neural and graph networks

[45] Physics-Informed Graph Neural Network for Spatial-temporal Production Forecasting

[46] Rediscovering orbital mechanics with machine learning

[47] Discovering Symbolic Models from Deep Learning with Inductive Biases

[48] GeomGCL Geometric Graph Contrastive Learning for Molecular Property Prediction

[49] Motif-based Graph Self-Supervised Learning for Molecular Property Prediction

[50] Rethinking Tokenizer and Decoder in Masked Graph Modeling for Molecules

[51] MoCL Data-driven Molecular Fingerprint via Knowledge-aware Contrastive Learning from Molecular Graph

[52] Self-supervised Graph-level Representation Learning with Local and Global Structure

[53] Graph Foundation Models

[54] GPS++ An Optimised Hybrid MPNN Transformer for Molecular Property Prediction

[55] Attending to Graph Transformers

[56] Towards Training Billion Parameter Graph Neural Networks for Atomic Simulations

[57] MuseGraph Graph-oriented Instruction Tuning of Large Language Models for Generic Graph Mining

[58] GraphEdit Large Language Models for Graph Structure Learning

[59] MolE a molecular foundation model for drug discovery

[60] Graph Meets LLMs Towards Large Graph Models

[61] Investigating Transfer Learning in Graph Neural Networks

[62] Structured Multi-task Learning for Molecular Property Prediction

[63] Molecular Representation Learning via Heterogeneous Motif Graph Neural Networks

[64] Learning Invariant Representations of Graph Neural Networks via Cluster Generalization

[65] Causal Attention for Interpretable and Generalizable Graph Classification

[66] The Heterophilic Snowflake Hypothesis: Training and Empowering GNNs for Heterophilic Graphs

[67] SizeShiftReg a Regularization Method for Improving Size-Generalization in Graph Neural Networks

[68] Efficient and Accurate Physics-aware Multiplex Graph Neural Networks for 3D Small Molecules and Macromolecule Complexes

[69] AnyGraph: Graph Foundation Model in the Wild

[70] A Molecular Multimodal Foundation Model Associating Molecule Graphs with Natural Language

[71] SciAgents: Automating scientific discovery through multi-agent intelligent graph reasoning

[72] Large Language Models are Effective Priors for Causal Graph Discovery

[73] LLM and GNN are Complementary: Distilling LLM for Multimodal Graph Learning

[74] The Impact of Large Language Models on Scientific Discovery a Preliminary Study using GPT-4

[75] GraphRNN Generating Realistic Graphs with Deep Auto-regressive Models

[76] Geometry-Complete Perceptron Networks for 3D Molecular Graphs

[77] Graph Neural Network with Local Frame for Molecular Potential Energy Surface

[78] Graph Transformation Policy Network for Chemical Reaction Prediction

[79] GraphDF A Discrete Flow Model for Molecular Graph Generation

[80] Constrained Graph Variational Autoencoders for Molecule Design

[81] Learning Deep Generative Models of Graphs

[82] Graph Machine Learning for Design of High-Octane Fuels

[83] Causal Discovery with Reinforcement Learning

[84] Graph Neural Network Bandits

[85] RetroGNN Approximating Retrosynthesis by Graph Neural Networks for De Novo Drug Design

[86] GFlowNets for AI-Driven Scientific Discovery

[87] Automated Scientific Discovery From Equation Discovery to Autonomous Discovery Systems

[88] GNNExplainer Generating Explanations for Graph Neural Networks

[89] GraphLIME Local Interpretable Model Explanations for Graph Neural Networks

[90] On Explainability of Graph Neural Networks via Subgraph Explorations

[91] Higher-Order Explanations of Graph Neural Networks via Relevant Walks

[92] ProtGNN Towards Self-Explaining Graph Neural Networks

[93] Encoding Concepts in Graph Neural Networks

[94] The Intelligible and Effective Graph Neural Additive Networks

[95] Generative Causal Explanations for Graph Neural Networks

[96] Reinforced Causal Explainer for Graph Neural Networks

[97] BAGEL A Benchmark for Assessing Graph Neural Network Explanations

[98] Deconfounding to Explanation Evaluation in Graph Neural Networks

[99] MotifExplainer a Motif-based Graph Neural Network Explainer

[100] MAGE: Model-Level Graph Neural Networks Explanations via Motif-based Graph Generation

[101] Generalizing Graph Neural Networks on Out-Of-Distribution Graphs

[102] Discovering Invariant Rationales for Graph Neural Networks

[103] DAG-GNN DAG Structure Learning with Graph Neural Networks

[104] Sample, estimate, aggregate A recipe for causal discovery foundation models

[105] Causal GNNs: A GNN-Driven Instrumental Variable Approach for Causal Inference in Networks

[106] Hierarchical Graph Neural Networks for Causal Discovery and Root Cause Localization

[107] Relating Graph Neural Networks to Structural Causal Models

[108] Counterfactual Learning on Graphs A Survey

[109] CF-GNNExplainer Counterfactual Explanations for Graph Neural Networks

[110] Learning and Evaluating Graph Neural Network Explanations based on Counterfactual and Factual Reasoning

[111] Debiasing Graph Neural Networks via Learning Disentangled Causal Substructure

[112] A General Framework for quantifying Aleatoric and Epistemic uncertainty in Graph Neural Networks

[113] Graph Posterior Network Bayesian Predictive Uncertainty for Node Classification

[114] Bayesian Graph Neural Networks with Adaptive Connection Sampling

[115] Graph Structure Learning with Interpretable Bayesian Neural Networks

[116] Bootstrapping Graph Convolutional Neural Networks for Autism Spectrum Disorder Classification

[117] Energy-based Out-of-Distribution Detection for Graph Neural Networks

[118] Uncertainty Aware Semi-Supervised Learning on Graph Data

[119] NOSMOG Learning Noise-robust and Structure-aware MLPs on Graphs

[120] EvenNet Ignoring Odd-Hop Neighbors Improves Robustness of Graph Neural Networks

[121] Probabilistic Weather Forecasting with Hierarchical Graph Neural Networks

[122] Causal Graph Neural Networks for Wildfire Danger Prediction

[123] Efficient Probabilistic Logic Reasoning with Graph Neural Networks

[124] On GNN explanability with activation rules

[125] KAN 2.0: Kolmogorov-Arnold Networks Meet Science

[126] Design Your Own Universe A Physics-Informed Agnostic Method for Enhancing Graph Neural Networks

[127] Molecular Graph Convolutions Moving Beyond Fingerprints

[128] Analyzing Learned Molecular Representations for Property Prediction

[129] Does GNN Pretraining Help Molecular Representation

[130] Deep Surrogate Docking Accelerating Automated Drug Discovery with Graph Neural Networks

[131] SS-GNN A Simple-Structured Graph Neural Network for Affinity Prediction

[132] Reliable Graph Neural Networks for Drug Discovery Under Distributional Shift

[133] Scalable deeper graph neural networks for high-performance materials property prediction

[134] ADA-GNN Atom-Distance-Angle Graph Neural Network for Crystal Material Property Prediction

[135] Structure-based out-of-distribution (OOD) materials property prediction a benchmark study

[136] Formula graph self-attention network for representation-domain independent materials discovery

[137] CHILI Chemically-Informed Large-scale Inorganic Nanomaterials Dataset for Advancing Graph Machine Learning

[138] GemNet-OC Developing Graph Neural Networks for Large and Diverse Molecular Simulation Datasets

[139] A Multimodal Graph Neural Network Framework of Cancer Molecular Subtype Classification

[140] MuSe-GNN Learning Unified Gene Representation From Multimodal Biological Graph Data

[141] Explainable Multilayer Graph Neural Network for Cancer Gene Prediction

[142] Learning Unknown from Correlations Graph Neural Network for Inter-novel-protein Interaction Prediction

[143] SHINE SubHypergraph Inductive Neural nEtwork

[144] PRAGA: Prototype-aware Graph Adaptive Aggregation for Spatial Multi-modal Omics Analysis

[145] SlideGraph+ Whole Slide Image Level Graphs to Predict HER2Status in Breast Cancer

[146] Latent-Graph Learning for Disease Prediction

[147] Deep Learning of Causal Structures in High Dimensions

[148] Benchmarking Graph Neural Networks

[149] TUDataset A collection of benchmark datasets for learning with graphs

[150] DrugOOD Out-of-Distribution (OOD) Dataset Curator and Benchmark for AI-aided Drug Discovery – A Focus on Affinity Prediction Problems with Noise Annotations

[151] Pitfalls of Graph Neural Network Evaluation

[152] A Fair Comparison of Graph Neural Networks for Graph Classification

[153] Are we really making much progress Revisiting, benchmarking, and refining heterogeneous graph neural networks

[154] Quantifying Explainers of Graph Neural Networks in Computational Pathology

[155] GraphFramEx Towards Systematic Evaluation of Explainability Methods for Graph Neural Networks

[156] DISCOVERYWORLD: A Virtual Environment for Developing and Evaluating Automated Scientific Discovery Agents

[157] Relational inductive biases, deep learning, and graph networks

[158] Trustworthy Graph Neural Networks Aspects, Methods and Trends